

OPTIMAL PAGE REPLACEMENT

```
GNU nano 8.7                                opc.c                                Modified
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n = 12;
    int faults = 0, hits = 0;
    int i,j,k,pos,found,far;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {
        found = 0;
        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                hits++;
                found = 1;
                break;
            }
        }
        if(!found) {
            pos = -1;
            far = i;
            for(j=0;j<3;j++) {
                int next = -1;
                for(k=i+1;k<n;k++) {
                    if(frames[j] == pages[k]) {
                        next = k;
                        break;
                    }
                }
            }
        }
    }

    printf("\noptimal Page Replacement\n");
    printf("Page Faults = %d\n", faults);
    printf("Page Hits = %d\n", hits);

    printf("Hit Percentage = %.2f%%\n", (float)hits/n*100);
    printf("Fault Percentage = %.2f%%\n", (float)faults/n*100);

    return 0;
}
```

```
GNU nano 8.7                                opc.c                                Modified
    int next = -1;
    for(k=i+1;k<n;k++) {
        if(frames[j] == pages[k]) {
            next = k;
            break;
        }
    }
    if(next == -1) {
        pos = j;
        break;
    }
    if(next > far) {
        far = next;
        pos = j;
    }
    if(pos == -1)
        pos = 0;
    frames[pos] = pages[i];
    faults++;
}

printf("\noptimal Page Replacement\n");
printf("Page Faults = %d\n", faults);
printf("Page Hits = %d\n", hits);

printf("Hit Percentage = %.2f%%\n", (float)hits/n*100);
printf("Fault Percentage = %.2f%%\n", (float)faults/n*100);

return 0;
}
```

```
athar@THOR MSYS ~  
$ nano opc.c  
athar@THOR MSYS ~  
$ gcc opc.c -o opc  
athar@THOR MSYS ~  
$ ./opc  
Optimal Page Replacement  
Page Faults = 7  
Page Hits = 5  
Hit Percentage = 41.67%  
Fault Percentage = 58.33%  
athar@THOR MSYS ~  
$
```